

AI DATA CENTER INFRASTRUCTURE

A Comprehensive Investment Research Report

Power, Photonics, Silicon, and the Software Stack

Prepared for	Clearvision Ventures, Menlo Park, California
Prepared by	Vince Yonesu Blessing Uche Yabes Bolas Hutajulu Ireen Mothello Johanes Paulus
Institution	University of San Francisco, School of Management
Program	Master of Science in Entrepreneurship and Innovation
Date	April 28, 2026
Classification	Confidential For Internal Use Only

This report is the product of 11.5 weeks of concurrent, independent research by five graduate students from the University of San Francisco completing practical training internships at Clearvision Ventures. Each researcher investigated a distinct domain of AI data center infrastructure. The findings have been compiled here as a single, unified investment analysis.

Table of Contents:

<i>Executive Summary</i>	3
<i>Introduction</i>	4
The Structural Reset	4
Research Approach and Scope	4
Gallium Nitride: The Near-Term Displacer	5
Key Milestones: 2025 Through April 2026.....	5
High-Temperature Superconductors: The Long-Horizon Solution	6
GaN and HTS as a Complementary Power Stack	7
The China Factor: Gallium and the November 2026 Catalyst	8
Investment Recommendations: Power Infrastructure	8
Five Investment Themes for Constraint Relief	9
Waste-Heat Recovery and Thermal Reuse.....	10
GPU Workload Orchestration and Power Efficiency Software	10
Advanced Liquid Cooling: Two-Phase Systems.....	10
Advanced Thermal Materials and Chip-Level Cooling	11
AI-Native Data Center Intelligence and Operations	11
Where the Value Is Shifting	12
The Recommended Category: Vendor-Neutral AI Chip Optimization	12
The State of the Market	14
Investment Framework: Three Layers with Highest Conviction	15
The Five-Layer Software Stack	16
Layer 1: Infrastructure Software	16
Layer 2: Management and Operations	16
Layer 3: AI and ML Workload Software	17
Layer 4: Security Software	17
Layer 5: Observability and Monitoring.....	17
Top Three Investment Categories for 2025 Through 2027	17
Emerging Whitespaces	18
Investment Screening Criteria	18
Energy as the Central Constraint	19
The Regulatory Tailwind Is Real and Accelerating	19
The Investment Window Is Closing, Not Opening	20
<i>Consolidated Investment Recommendations</i>	21
<i>Conclusion and Strategic Recommendations</i>	23
<i>Appendices</i>	25
Appendix A: Key Market Data Summary	25
Appendix B: Dated Catalysts and Investment Timeline	25
Appendix C: Investment Screening Checklist	26
Appendix D: Sources and References	27

Executive Summary

The data center industry is not just growing. It is being restructured from the inside out, forced by AI workloads that have made the assumptions behind the old infrastructure model obsolete. Rack power density has climbed from roughly 10 kW in a conventional enterprise deployment to well over 300 kW in leading AI clusters today. Vertiv projects that figure will reach 1 MW per rack by 2029. The global electrical and software architecture built over the past three decades was not designed for this, and it cannot be patched to handle it.

This report compiles the findings of five graduate researchers from the University of San Francisco who spent 11.5 weeks conducting concurrent, independent investigations into distinct domains of AI data center infrastructure during their practical training placement at ClearVision Ventures. The research covered GaN semiconductors and high-temperature superconductors as power infrastructure solutions; cooling systems, waste heat recovery, and the physical constraints reshaping facility design through 2036; silicon efficiency and the venture opportunity in the compute optimization layer; photonic interconnects and the copper-to-light transition underway in AI cluster networking; and the data center software stack, including where AI complexity has outpaced incumbent vendor capabilities.

What is striking is how consistently the five research streams point to the same underlying conclusion. The infrastructure constraints binding AI buildout today are not demand problems. They are physics and architecture problems. Copper cannot move data fast enough at AI speeds without burning energy on signal regeneration. Silicon inefficiency means expensive GPU clusters sit idle or underutilized. Legacy software was designed for air-cooled enterprise racks, not facilities running 100,000 sensors and liquid-cooled AI workloads. The power grid cannot expand fast enough to meet demand, and copper cable management is physically untenable at megawatt rack densities.

The five investment themes emerging from this research are: GaN-based power conversion enabling 800V HVDC architectures; high-temperature superconducting cables for near-lossless power delivery at the campus and transmission layer; photonic interconnects replacing copper inside AI clusters; vendor-neutral silicon efficiency software maximizing the productivity of installed compute; and AI-native data center intelligence covering energy management, cooling optimization, and regulatory compliance automation. The investment window across all five is compressing. Commercial validation has happened. Early-stage valuations remain accessible today, but not for long.

Introduction

The Structural Reset

The shift from CPU-centric cloud workloads to GPU-heavy AI training and inference has not added new demand on top of an existing system. It has broken the system. A conventional server rack tops out at 30 kW. An NVIDIA Blackwell GB200 NVL72 runs at 120 kW. The forthcoming Vera Rubin generation pushes beyond 300 kW per rack. These are not incremental changes. They are order-of-magnitude increases happening inside a physical plant built for a fundamentally different regime.

The numbers frame the scale of the problem. Worldwide data center capital expenditure increased 57 percent in 2025, reaching over \$420 billion among the top hyperscalers, with full-year 2026 forecasted to surpass \$1 trillion (Dell'Oro Group, March 2026). Global data center electricity demand sat at approximately 415 TWh in 2024 and is projected to reach around 945 TWh by 2030 (IEA). In the United States, Morgan Stanley projects a 49 GW power shortfall by 2028. Half of planned 2026 US data center builds are already delayed or canceled, with electrical component lead times cited as the root cause (Bloomberg, Sightline Climate).

This is the backdrop against which each of the five research areas in this report was conducted. Power semiconductors, superconducting cables, photonic interconnects, silicon efficiency software, and data center operations are not the same problem. But they share a common driver and a common opportunity profile for ClearVision Ventures: each addresses a specific, documented bottleneck that stands between the capital being committed to AI infrastructure and the operational AI capacity that capital is supposed to create.

Research Approach and Scope

Over 11.5 weeks, the five researchers drew on primary and secondary sources published between 2024 and April 2026. Each research stream is intentionally focused: GaN and HTS covers power conversion and delivery; cooling and thermal covers facility-level heat management and grid constraint; silicon covers the efficiency layer above deployed accelerators; photonics covers the copper-to-light transition in cluster interconnect; and software maps the five-layer data center stack for venture return potential. The researchers worked independently and then contributed their findings to this compiled document. Where the streams converge, that is noted explicitly. Where they address different horizons, that context is preserved.

SECTION 1 | POWER INFRASTRUCTURE: GaN AND HIGH-TEMPERATURE SUPERCONDUCTORS

The Power Crisis: Why It Is Happening Now

The power crisis facing data centers arrived suddenly rather than gradually. The shift to GPU-heavy AI workloads-imposed physics constraints that copper-based electrical architecture was not built to handle. The I-squared-R law makes this irreversible: power loss on a backplane scale with current squared. At 12V and 300 kW, the current flowing through copper conductors reaches roughly 25,000 amperes. The resistive heat generated at that scale forces redundant cable runs, expands cooling loads, and consumes floor space before a single computation begins. Raising voltage is the only physical escape. At 800V, the same 300 kW requires just 375 amperes, cutting copper mass and resistive losses by more than 99 percent.

The copper supply problem compounds the physics problem. Wood Mackenzie reported a 304,000-tonne copper deficit in 2025 that widened into 2026. UBS estimates a 400,000-tonne shortfall in 2026 as mines in Chile, Peru, and Indonesia face operational disruptions. S&P Global projects a 10 million metric ton shortfall by 2040. A single 100 MW data center absorbs 27 to 33 tonnes of copper before any upstream grid work begins. There is no version of this problem that more copper resolves.

Even unlimited copper supply would not fix the grid access problem. New US transmission lines take 10 or more years to permit. New substations take 3 to 7 years. Gartner estimated in November 2024 that 40 percent of AI data centers will be operationally constrained by power availability by 2027. Two materials are positioned to resolve this from the inside: Gallium Nitride at the conversion layer within the data center, and High-Temperature Superconductors at the delivery layer between the grid and the building.

Gallium Nitride: The Near-Term Displacer

GaN is not a future technology. It is already shipping inside production AI servers today. What has changed in 2025 and 2026 is the scale of deployment, the standardization of the architecture it enables, and the maturity of the manufacturing base behind it. The transition from silicon to GaN in data center power conversion is no longer a question of whether but of how fast.

The fundamental advantage is physical. GaN has a bandgap of 3.4 eV compared to silicon's 1.1 eV, meaning it sustains much higher electric fields before breakdown. Combined with high electron mobility, GaN transistors switch at frequencies into the megahertz range, roughly 10 to 20 times faster than silicon's typical 50 to 100 kHz. Higher switching frequency allows smaller passive components, directly reducing the physical footprint of conversion stages and enabling the high power density that AI server designs require. At system level, GaN achieves power densities approaching and exceeding 2 kW per cubic inch, compared to roughly 300 to 600 W per cubic inch for silicon.

The architectural centerpiece of the GaN story is the move from 48V distribution to 800V HVDC. NVIDIA formalized this with its Kyber architecture, which distributes 800V DC from a sidecar power rack to compute racks, enabling direct 800V-to-point-of-load conversion inside the server tray. Silicon MOSFETs cannot operate efficiently at 800V in the thermal and space constraints of AI server trays. GaN is the only practical choice at this conversion stage.

Key Milestones: 2025 Through April 2026

Several milestones across this period have retired the three legitimate barriers that historically slowed GaN adoption in data centers: cost versus silicon, gate drive complexity, and qualification risk for hyperscale procurement.

At NVIDIA GTC in March 2026, Navitas debuted a DC-DC power delivery board enabling direct 800V-to-6V conversion in a single power stage, eliminating the traditional 48V intermediate bus converter stage entirely. The board delivers 96.5 percent peak efficiency at full load, operates at 1 MHz switching frequency, and achieves 2,100 W per cubic inch power density. On the same day, Texas Instruments unveiled a complete 800V DC power architecture demonstrating only two conversion stages from 800V to processor power, with 97.6 percent peak efficiency and over 2,000 W per cubic inch. One of the world's largest semiconductor

companies confirming that the 800V GaN architecture works at system level in production-ready form is not a minor data point.

Infineon's shipment of the first 300 mm GaN wafer production run in 2024 unlocked cost curves comparable to mature silicon processes. A 12 kW GaN-based power supply can now be manufactured at the same footprint as a legacy 3.3 kW silicon unit, effectively quadrupling power density at similar cost. GaN part pricing has dropped over 50 percent in the past year. In April 2026, Intel Foundry Services announced the world's thinnest gallium nitride chiplet, reducing the silicon substrate to just 19 micrometers and achieving monolithic integration of GaN power transistors with silicon-based digital logic circuits for the first time. onsemi in March 2026 introduced vertical GaN semiconductors handling up to 1,200V with energy losses reduced by almost 50 percent. The ecosystem acceleration is broad and competitive across all major players.

Company	Product or Milestone	Key Metric	Date
Navitas	800V-to-6V single-stage DC-DC board	96.5% peak efficiency, 2,100 W/in ³	March 2026
Texas Instruments	Complete 800V DC architecture	97.6% peak efficiency, 2,000 W/in ³	March 2026
Navitas	10 kW all-GaN 800V-to-50V platform	98.5% peak efficiency, 2.1 kW/in ³	February 2026
Infineon	300mm GaN wafer production	12 kW unit at footprint of legacy 3.3 kW Si unit	2024
onsemi	Vertical GaN on GaN-on-GaN substrate	50% lower losses, handles 1,200V, 3x smaller form factor	March 2026
Intel Foundry	World's thinnest GaN chiplet	19 micrometer substrate, first GaN and Si monolithic integration	April 2026
Renesas	Gen IV+ GaN FETs, 650V	14% lower on-resistance, silicon-compatible gate drive	July 2025

High-Temperature Superconductors: The Long-Horizon Solution

HTS operates on a different timeline and solves a different part of the power crisis. Where GaN optimizes the conversion of power inside the data center, HTS eliminates resistive loss in the delivery of power to it, specifically the transmission layer between the grid substation and the building. A superconducting cable carries current with effectively zero electrical resistance, meaning zero resistive heat generation and no voltage drop regardless of current magnitude. A single REBCO-based HTS cable can carry the same power as dozens of copper cable runs in a fraction of the space.

A study published in ScienceDirect documented residual loss of just 1.7 percent in an HTS transmission system at 100 MW scale, compared to the 3 to 5 percent resistive loss of a copper baseline. MetOx's analysis from September 2025 found HTS cables offer more than 80 percent reduction in cable footprint for equivalent power delivery. At campus-scale data centers receiving 300 MW, that difference is measured not in cable trays but in substation footprint, right-of-way requirements, permitting timelines, and civil construction cost.

The most significant basic research development in HTS in three decades arrived in March 2026, when researchers at the University of Houston achieved a transition temperature of 151 Kelvin under ambient pressure, published in the Proceedings of the National Academy of Sciences. This surpassed the previous 133 K record that had stood since 1993. The technique, called pressure quenching, applies intense pressure to enhance superconducting properties, cools the material, then rapidly releases the pressure to lock in the enhanced state. Every degree the transition temperature rises means less cryogenic energy needed in a deployed cable, which directly improves the economics of data center HTS deployments.

On the commercial side, VEIR demonstrated 3 MW of power transmitted through a single low-voltage HTS cable in a simulated data center environment in November 2025, the world's first demonstration of this kind. Microsoft confirmed factory testing of the full 3 MW cable system and published a technical overview of the world's first HTS-powered rack. Microsoft invested in VEIR as part of its \$75 million Series B alongside National Grid and Munich Re Ventures. VEIR is targeting full commercial launch in 2027.

The manufacturing tailwind from the fusion energy sector is a cross-sector catalyst that most data center analysts are not tracking. Approximately \$10 billion in private capital has flowed into fusion energy companies, the majority of which use REBCO tape as a core component of their high-field magnets. Top producers now exceed 3,000 km of tape per year. REBCO tape costs \$15 to \$30 per meter today, against a target of approximately \$10 per meter for data center economic viability, with fusion capex as the primary cost-reduction driver over the next three to five years.

GaN and HTS as a Complementary Power Stack

No published industry report as of this writing models GaN and HTS as a single complementary power stack. Most analysis treats them as separate categories: one a semiconductor story, the other a materials science story. That framing misses the most important structural point. They solve different layers of the same problem and their value compounds when deployed together.

The architecture is straightforward. HTS cable carries power from the grid substation to the data center building with near-zero resistive loss, with one cable replacing dozens of copper runs delivering 300 MW through existing corridors. At the building boundary, power enters bus and PDU infrastructure at 48V to 800V HVDC. GaN conversion stages at the rack level then handle all remaining voltage conversions from 800V down to the 50V, 12V, or sub-1V levels that GPUs and compute substrates require, at greater than 97 percent efficiency. HTS eliminates loss at the macro scale. GaN optimizes what remains at the micro scale.

There is also a structural cross-technology convergence that almost no analyst is modeling. NVIDIA's Vera Rubin NVL72 requires 100 percent liquid cooling with no air-cooled configuration. Hyperscale data centers are now installing liquid cooling infrastructure at massive scale driven by AI compute density. HTS cables also require liquid cooling, typically liquid nitrogen. The thermal infrastructure being built for AI compute

creates exactly the conditions HTS needs to deploy. The data center industry is, in effect, building the support system for HTS deployment as a side effect of solving a separate problem.

HTS eliminates resistive loss at the transmission layer. GaN optimizes the remaining conversion stages at the rack. No existing industry report models both as a complementary stack. The convergence of these two technologies one mature and deploying now, one commercializing in 2026 to 2027 is the structural power thesis for the next decade of AI infrastructure.

The China Factor: Gallium and the November 2026 Catalyst

China controls approximately 98 percent of global primary gallium supply, the raw material every GaN semiconductor device requires. In 2023, China introduced export restrictions on gallium and germanium. By May 2025, prices had reached \$687 per kilogram, up 150 percent from pre-restriction levels. China subsequently suspended the full ban, with the suspension currently scheduled to expire on November 27, 2026.

When that suspension expires, Beijing can reinstate the full export ban with no advance notice. Every Western GaN manufacturer without domestic gallium supply becomes structurally more exposed the moment it does. The market is not currently pricing this risk because the suspension is still in effect. That is a mispricing with a known expiry date attached to it.

MTM Critical Metals, a Texas-based startup, has developed a patent-protected process for recovering gallium from industrial scrap and is the only US gallium production facility independent of Chinese raw material inputs, with a binding offtake agreement with Indium Corporation and operations launched in early 2026. The Navitas-GlobalFoundries long-term manufacturing partnership announced in 2025 represents a parallel supply chain response, with US domestic GaN manufacturing availability targeted for late 2026.

Investment Recommendations: Power Infrastructure

Company	Stage	Investment Thesis	ClearVision Fit
MetOx International	Series B Ext. ~\$126M raised, \$80M DOE funding	Manufactures Xeus REBCO HTS wire. Pick-and-shovel play on the entire HTS category. Every HTS cable deployment requires MetOx wire.	Best current fit. Pre-revenue, post-government validation. Strong downside protection from federal backing.
MTM Critical Metals	Early stage, operations live 2026	Domestic gallium recovery from industrial scrap. Only US facility independent of Chinese inputs. Binding offtake with Indium Corporation.	Direct November 27 ban-expiry catalyst. ClearVision sweet spot on stage and check size.

Amber Semiconductor	Series C, \$30M closed March 2026	PowerTile vertical power delivery beneath AI processors. 85% loss reduction. Shipping to hyperscaler customers Q3 2026.	Live entry window with named delivery milestone. GaN application-layer reference company.
VEIR	Series B, ~\$117M raised, ~\$170M valuation	Complete HTS power delivery systems for data centers. World-first 3 MW OCP demo. 2027 commercial launch target.	Above lead range. Co-invest or Series C participation as 2027 commercial launch approaches.
SuperNode	~\$38M raised, pre-commercial	Next-generation superconducting cable systems. Demonstrated 500 MW equivalent through 30m cable. CERN collaboration.	Early-relationship opportunity before institutional capital crowds the round.

SECTION 2 | COOLING, THERMAL MANAGEMENT, AND THE DATA CENTER OF 2036

The Physical Constraint System

The data center industry is entering a structural transition in which AI workloads are turning power, cooling, grid access, and regulation into the primary limits on growth. Demand is rising faster than efficiency improvements can absorb. Global data center electricity consumption is projected to triple by 2035, while US demand could reach 325 to 580 TWh by 2028. Rack density has moved from ordinary enterprise-server levels to AI-infrastructure levels, with Blackwell-class racks already exceeding 100 kW and next-generation systems expected to reach several hundred kW by 2027.

Air cooling is reaching its physical limit because it cannot remove heat fast enough at modern GPU densities. But cooling is part of a broader power-and-thermal constraint system rather than an isolated problem. Grid access is now the deeper expansion constraint: interconnection timelines in major markets stretch beyond four years, while transmission build-out has slowed just as AI demand accelerates. The strongest investment opportunities are therefore not in building more data center capacity. They are in the enabling technologies that allow high-density facilities to receive power, remove heat, stay compliant, and operate reliably.

It is also worth acknowledging the Jevons Paradox dynamic at work here. Efficiency gains are not reducing total energy demand. AI follows the same historical pattern as every prior efficiency-improving technology: cheaper and more efficient computation expands usage faster than it saves energy. Agentic AI will intensify this by replacing bursty inference with continuous, always-on workloads that reduce thermal recovery windows and require real-time operational intelligence. The energy problem does not go away as hardware gets more efficient. It becomes more complex.

Five Investment Themes for Constraint Relief

Waste-Heat Recovery and Thermal Reuse

The strongest near-term investment theme is waste-heat recovery, and it earns that position specifically because demand is increasingly mandated rather than discretionary. The EU Energy Efficiency Directive entered force in January 2025. Germany's 20 percent waste-heat reuse requirement carries an enforcement date of July 2028. The EU Data Centre Energy Efficiency Package of April 2026 extends these obligations further. This changes the adoption risk profile because operators must respond to compliance pressure, not just economic incentive.

The economics are improving independently of regulation. A facility rejecting 50 MW of waste heat at 40 to 60 degrees Celsius into a district heating network at 20 to 40 euros per MWh can convert what was an environmental liability into a revenue stream. Companies building credible positions here include Karman Industries, which is developing CO₂-based heat-processing systems for AI-era thermal loads; Qpinch, which upgrades low-grade industrial waste heat into useful process heat; and Climeon, which provides a commercial reference model for low-temperature heat-power recovery.

GPU Workload Orchestration and Power Efficiency Software

Many colocations operate at only 30 to 50 percent GPU utilization, and even best-in-class hyperscalers struggle to sustain utilization above 60 to 70 percent. A Blackwell rack carrying \$4 million in hardware at 40 percent utilization can leave millions of dollars of deployed capital underused while its cooling system still runs for full rated thermal design power. Dynamic voltage and frequency scaling research demonstrates 19 to 32 percent energy savings from software-layer optimization alone.

NVIDIA's approximately \$700 million acquisition of Run:AI validates the strategic value of this layer and sets a meaningful exit benchmark. The shift from power usage effectiveness to tokens per watt as the key operating metric creates a first-mover advantage for companies building products around the new metric before the hardware ecosystem catches up. Companies showing momentum here include Crusoe, through energy-first AI infrastructure and cloud compute, and Giga ML, through enterprise LLM deployment and fine-tuning tools. Defensibility depends on integration depth and multi-vendor compatibility rather than algorithmic novelty alone.

Advanced Liquid Cooling: Two-Phase Systems

Air cooling is now a minority position for the first time in the industry's recorded history, and roughly 95 percent of the liquid-cooling transition still lies ahead. Two-phase liquid cooling earns priority because it addresses the density tier that single-phase systems may struggle to reach, including Vera Rubin systems above 300 kW and Rubin Ultra systems projected near 600 kW, while offering waterless dielectric operation and higher inlet-water tolerance that can expand free-cooling hours.

The strategic acquirer market is active and pricing is meaningful. Ecolab agreed to acquire CoolIT for approximately \$4.75 billion, representing 29x estimated next-12-month EBITDA. Leading companies include Accelsius, which closed a \$65 million Series B led by Johnson Controls with Legrand participating and has deployments linked to Equinix Ashburn and Telehouse London; Submer, providing immersion-cooling systems aligned with European sustainability pressure; and Iceotope, focusing on precision liquid cooling compatible with existing rack infrastructure.

Advanced Thermal Materials and Chip-Level Cooling

As rack-level thermal design power climbs from 163 kW today toward 300 to 600 kW by 2027, the frontier of cooling is moving progressively closer to the chip. Materials science and microfluidic engineering require specialized expertise, proprietary manufacturing processes, and multi-year hyperscaler validation cycles, creating genuine moats that incumbent HVAC suppliers cannot easily replicate. The EU's F-Gas Regulation adds a regulatory tailwind, while ARPA-E's COOLERCHIPS programme, targeting cooling energy below 5 percent of total IT load, validates the engineering trajectory.

Companies worth watching include Corintis, which has raised Series A and strategic capital for microfluidic chip cooling with customer validation from major technology buyers; ZutaCore, offering waterless two-phase direct-to-chip cooling with heat-reuse potential; and GRC, demonstrating a volume deployment path for immersion-based cooling systems.

AI-Native Data Center Intelligence and Operations

Legacy DCIM tools were designed for air-cooled enterprise facilities with static thermal profiles. AI-era data centers require software built for 100,000-plus sensors, liquid-cooling loops with real-time fluid-chemistry monitoring, GPU clusters generating continuous thermal loads, and grid-interactive demand response. A platform that can manage cooling setpoints, predict cold-plate fouling before it causes GPU throttling, schedule workloads according to thermal headroom, and translate compute decisions into utility demand-response signals creates compounding switching costs because every added data stream improves model accuracy.

The February 2026 validation of grid-interactive AI compute shows that demand response is technically real and strengthens the commercial pitch by turning the value proposition from cost reduction into potential revenue generation. The strongest entry strategy is to begin with mid-market colocation operators and enterprise data centers before expanding upmarket, given the long enterprise infrastructure software sales cycle.

Theme	Conviction	Stage	Check Size	Deploy
Waste-Heat Recovery	Highest	Series A/B	\$5M to \$15M	Now
GPU Orchestration Software	Very High	Seed to Series A	\$3M to \$10M	Now
Two-Phase Liquid Cooling	High	Series B/C	\$10M to \$20M	2026 to 2027
Thermal Materials and Chip Cooling	High	Series B	\$8M to \$15M	2026 to 2028
AI-Native DCIM and Operations	Good	Series A Lead	\$5M to \$12M	Source now

The data center of 2036 will not be defined by how large it is, but by whether it was designed from the outset around the power, thermal, water, and regulatory constraints that AI workloads impose. Value accrues disproportionately to technologies that relieve those bottlenecks rather than those that simply scale with demand.

SECTION 3 | SILICON EFFICIENCY AND THE COMPUTE OPTIMIZATION LAYER

Silicon as a Strategic Control Point

Silicon has moved from a background hardware input to a strategic control point in modern data centers. AI workload growth has made the economics of infrastructure dependent on how effectively accelerators, memory, interconnect, and power operate together, rather than on the raw performance of any individual component. For investors, the implication is clear: value is shifting from the chip alone to the efficiency of the full compute system. The IEA estimates global data center electricity demand at about 415 TWh in 2024, projected to reach roughly 945 TWh by 2030. US data centers consumed around 180 TWh in 2024 and are expected to drive a significant share of demand growth through the rest of this decade.

AI has made power, cooling, and interconnect commercial constraints rather than technical afterthoughts. As clusters grow, bottlenecks increasingly emerge outside the processor itself, which creates urgency for data-movement and utilization tools. Large incumbents are strongest at the core silicon layer but weaker across heterogeneous optimization workflows. As customers run mixed fleets, independent optimization layers become more valuable than single-vendor tools. The best venture opportunities sit close enough to silicon to capture economic value but far enough away to avoid foundry-scale capital risk.

Where the Value Is Shifting

ClearVision should be cautious about direct bets on full-stack general-purpose AI accelerator companies. Capital requirements, long qualification cycles, and strong incumbent response all weaken the probability-adjusted venture case. The better question is where a startup can influence silicon economics without carrying the full burden of manufacturing and platform replacement.

The most attractive wedge is improving the productivity of compute already deployed in the market. Customers care about latency, throughput, utilization, deployment flexibility, and power consumption. A startup that measurably improves those outcomes has a clear commercial story, particularly as fleets become more heterogeneous. Mixed hardware environments increase the value of vendor-neutral optimization software that improves performance without forcing a full platform rewrite.

Inference is the strongest initial wedge because it links technical performance directly to operating economics. Customers can measure the value of inference optimization quickly through latency, utilization, and token cost. A startup that wins in inference can later expand into orchestration, model routing, and broader workload optimization without starting with an overly broad product scope.

The Recommended Category: Vendor-Neutral AI Chip Optimization

The recommended investment category for ClearVision in this domain is vendor-neutral AI chip optimization infrastructure focused initially on inference efficiency, workload orchestration, and model-serving economics. In practical terms, this means a company that sits between AI models and heterogeneous accelerator fleets to improve performance, utilization, and cost efficiency.

The return case is credible because customer economics are strong. If a platform reduces token-serving cost or lifts useful output enough to save a customer several million dollars annually, it can command a high-value software contract while still delivering clear ROI. An illustrative scenario is 20 to 30 meaningful customers at roughly \$2 million to \$4 million of annual contract value, implying \$40 million to \$100 million of recurring revenue without requiring universal market dominance. If a company reaches approximately \$75 million in recurring revenue with strong retention, outcomes in the high hundreds of millions of dollars are realistic. The startup does not need to displace NVIDIA or any hyperscaler to succeed. It only needs to own a meaningful efficiency layer in a growing part of the AI infrastructure market.

Insight	Investment Implication
Value creation has shifted from chips to compute efficiency	Prioritize technologies improving delivered compute rather than theoretical performance in isolation
AI made power, cooling, and interconnect commercial constraints	Bottlenecks outside the processor create urgency for data-movement and utilization tools
Incumbents are weak across heterogeneous optimization workflows	As customers run mixed fleets, independent optimization layers become more valuable
Best opportunities are close to silicon but not in chip design	Optimization infrastructure offers a stronger balance of urgency, scalability, and capital efficiency
Timing is favorable as the market builds out	AI infrastructure investment remains elevated while architectures are still evolving, which is the most favorable phase for early-stage entry

SECTION 4 | PHOTONIC INTERCONNECTS: THE COPPER-TO-LIGHT TRANSITION

Why Copper Has Already Hit Its Limit

At 224 gigabits per second per lane, the speed required for modern AI interconnect, copper cables cannot transmit data more than one meter without active signal regeneration. This is not an engineering problem awaiting a clever solution. It is the skin effect: a fundamental property of electromagnetism that concentrates high-frequency electrical current at the conductor surface, increasing resistance and signal loss non-linearly. No copper engineering advancement changes this physics.

At the scale of an 18,000-GPU cluster, the retimer chips required to regenerate copper signals across a two-layer network add approximately 480 kilowatts of power consumption solely for moving data between chips, roughly 1.8 kilowatts per server rack. NVIDIA's CEO Jensen Huang made this explicit at GTC 2025: optical

interconnects now consume approximately 10 percent of total compute budget in a modern AI cluster. He cited this as the primary motivation for co-packaged optics development.

Copper electrical interconnects at AI speeds consume 55 picojoules of energy per bit moved. Co-packaged optical systems consume fewer than 5. At a hyperscale facility with one million active data connections, that difference produces annual electricity savings exceeding \$175 million. At 100,000 GPU scale, total annual operating savings from photonic adoption reach \$250 million per year. The capital premium for photonic interconnect over copper is approximately 1.9 percent of total cluster cost, a premium that pays for itself in seven months.

The State of the Market

On April 7, 2026, Semiconductor Engineering reported from the Optical Fiber Communications Conference that all high-bandwidth AI data center interconnects will be optical within five years. On the same day, the UALink Consortium released version 2.0 of its open interconnect standard, backed by every major technology company except NVIDIA. These are declarations of commercial intent by the companies that build and run AI infrastructure, not analyst forecasts.

Broadcom shipped the first commercial co-packaged optics platform in October 2025. NVIDIA's Quantum-X InfiniBand switch with co-packaged optics began shipping in early 2026. Meta confirmed that co-packaged optics is more reliable than pluggable transceivers after one million link-hours of high-temperature testing without a single failure. The technology has crossed the line from prototype to production.

The supply chain has a structural bottleneck that money alone cannot rapidly resolve. Silicon cannot generate light. Every photonic system depends on indium phosphide lasers. In 2025, global demand for indium phosphide devices reached 2 million units while production capacity was only 600,000, a 70 percent supply-demand gap. NVIDIA has pre-allocated EML laser capacity at both Lumentum and Coherent, the two dominant suppliers, pushing industry-wide lead times beyond 2027. Companies unable to secure laser allocation face infrastructure delays regardless of budget.

Player	Role	Why It Matters
NVIDIA	Ecosystem controller	CUDA, NVLink, NVSwitch, and NCCL form an interlocking proprietary stack. NVLink Fusion opens a partial route to market for photonic component companies.
Broadcom	Structural winner	\$46B AI revenue in 2026, up 134 percent. Ships the industry's first 102.4 Tbps co-packaged optics switch. Benefits regardless of which GPU vendor dominates.
Marvell and Celestial AI	Exit validation	\$3.25B acquisition in February 2026. Definitive exit benchmark for photonic chip design companies.
Lightmatter	Deepest moat architecture	Passage 3D photonic interposer claims 114 Tbps per package by enabling I/O across the full chip surface. Commercial timeline 2027 to 2028.

Ayar Labs	Most validated startup	Demonstrated 100% CPO prototype rack at OFC 2026. TeraPHY optical engines on TSMC process. Natural acquisition target.
POET Technologies	Laser bottleneck solution	Optical Interposer integrates lasers, modulators, and electronics on a single chip. \$5.6M in Q3 2025 production orders. Foxconn partnership. 3.2T optical engine sampling H2 2026.
UALink Consortium	Open market catalyst	100+ members including AMD, Intel, Google, Microsoft, Meta. V2.0 released April 7, 2026. Creates a \$3B+ photonics market outside the NVIDIA ecosystem by 2030.

Investment Framework: Three Layers with Highest Conviction

Photonic chip design is the rarest engineering capability in the space. Design tools are still immature compared to electronic equivalents, and proprietary architectures with hyperscaler design wins command acquisition premiums. The Celestial AI acquisition at \$3.25 billion is the definitive exit benchmark. Target companies include Lightmatter, Ayar Labs, and POET Technologies.

Advanced packaging and integration is the hardest engineering problem in photonics. Assembling photonic chips, lasers, fibers, and electronics into working systems with sub-micron precision is something very few companies can do at production yield. Switching costs are high once a company is embedded in a customer's supply chain. The TSMC COUPE program and Broadcom's CPO platform are the benchmarks.

Laser materials innovation addresses the deepest structural constraint in the full stack. Any startup eliminating or reducing dependence on externally sourced indium phosphide lasers through on-chip integration or alternative materials addresses a bottleneck affecting every photonic product in the market. The 70 percent InP demand-supply gap and NVIDIA's lockup through 2027 define the urgency. POET Technologies is the most advanced commercial company working in this space today.

On the avoidance side, copper infrastructure, DSP-based transceiver module manufacturers, and any company whose primary value proposition is making copper work better at AI speeds represent depreciating technologies. Any company presenting only roadmap slides without foundry booking confirmation, packaging yield data from existing production, and named hyperscaler design wins should not pass ClearVision's screening threshold.

Period	Key Transition Milestones
2026	CPO in initial production at Broadcom and NVIDIA. Pluggable 1.6T transceivers becoming the baseline. UALink 2.0 specification released. InP laser shortage most acute.
2027	UALink silicon in commercial production. CPO scaling to 30 percent or more of cluster interconnect at tier-1 hyperscalers. Laser supply beginning to ease as 6-inch InP wafer production ramps.

2028	Optical circuit switching entering production for building-scale fabric. CPO packaging yield standards established. First fully optical AI training clusters in operation.
2030 and beyond	CPO as default interconnect for all new AI cluster builds. Copper retained only for intra-chip connections. Energy consumption per AI token down 70 percent or more versus 2025 baseline.

SECTION 5 | DATA CENTER SOFTWARE LANDSCAPE 2025 TO 2027

The Software Investment Thesis

The data center software stack is undergoing its most significant structural reset in a decade, driven by AI workloads that demand fundamentally different orchestration, observability, and energy management capabilities. Worldwide data center capital expenditure increased 57 percent in 2025, reaching over \$420 billion among the top hyperscalers, with full-year 2026 forecasted to surpass \$1 trillion (Dell'Oro Group, March 2026). Large hardware capital expenditure cycles create software pull-through, and AI complexity has outpaced existing vendor capabilities across multiple categories simultaneously. That creates a rare entry window for software-layer investment ahead of the hardware cycle settling.

The overarching thesis is that the AI-driven infrastructure supercycle is creating a structural software replacement cycle across data center operations. The most defensible, highest-return opportunities lie in categories where AI complexity has structurally outpaced incumbent vendor capabilities: AI-native DCIM and energy management, inference infrastructure optimization, and multi-cloud security automation. The software-defined data center market was projected at \$72.3 billion in 2024, growing at 30.4 percent CAGR to 2033 (Grand View Research, 2024).

The Five-Layer Software Stack

The data center software stack spans five distinct layers with materially different growth rates, competitive dynamics, and venture return profiles. Understanding where value accrues requires mapping each layer honestly, including where the opportunity has already closed.

Layer 1: Infrastructure Software

Kubernetes has won the container layer and is maintained as open-source by the CNCF. The VMware and Broadcom pricing disruption primarily benefits open-source alternatives like Proxmox rather than VC-backed startups. The container orchestration market sits at approximately \$1.7 billion growing at 31.8 percent CAGR to 2030, but the dominant players are open-source foundations. The only viable opportunity in Layer 1 is differentiated control-plane software above Kubernetes, not alternatives to it. Venture conviction here is low.

Layer 2: Management and Operations

AI-native DCIM is the highest-conviction whitespace in data center software. The DCIM market stands at approximately \$3.6 billion in 2025, projected to reach \$9.9 billion by 2031 at an 18.3 percent CAGR (Mordor Intelligence, 2025). The top five DCIM players hold only approximately 26 percent combined market share

(Global Market Insights, 2024). The incumbents, Schneider, Vertiv, and Eaton, are hardware-first companies with no structural incentive to build AI-native, hardware-agnostic software. The EU Energy Efficiency Directive plus US EPA reporting mandates create mandatory compliance infrastructure demand. One percent efficiency gain equals approximately \$200 million in annual savings across North America and Europe (Energize Capital, 2025). Venture conviction here is strong.

Layer 3: AI and ML Workload Software

MLOps is the fastest-growing defined data center software category at 28.9 percent CAGR from \$2.33 billion in 2025 to \$25.9 billion by 2034 (Fortune Business Insights, 2025). Inference optimization has no clear category winner. 2026 is the training-to-inference pivot year, and the Series A entry window is 12 to 18 months. Baseten raised \$75 million in its Series C in February 2025 led by IVP and Spark. Lambda raised \$1.5 billion in its Series E in November 2025 led by TWG Global. SageMaker, Azure ML, and Vertex AI compete only within single clouds, meaning multi-cloud-native inference orchestration is structurally independent from hyperscalers. Venture conviction here is the highest of any software layer.

Layer 4: Security Software

The data center security market stands at approximately \$17.9 billion in 2025, growing at 11.9 percent CAGR (Precedence Research, 2025). The sub-category to watch is AI compliance automation. The EU AI Act GPAI obligations entered effect in August 2025, with full enforcement for high-risk AI systems beginning August 2026. No incumbent GRC platform has purpose-built AI compliance modules for EU AI Act Article 9 risk management requirements, model cards, data provenance, bias audit trails, or inference endpoint monitoring. Cyber insurers are now requiring AI audit capabilities as prerequisites for coverage renewal. Venture conviction here is strong.

Layer 5: Observability and Monitoring

General observability has consolidated. Datadog reported \$3.4 billion in FY2025 revenue with approximately 32,700 customers. Cisco completed its \$28 billion acquisition of Splunk in March 2024, anchoring a further consolidation wave. The only viable entry is AI-native model observability covering token costs, hallucination rates, and model drift as LLM-specific metrics. This niche remains genuinely open. The general category does not. Venture conviction here is low except for narrow AI-native niches.

Top Three Investment Categories for 2025 Through 2027

Rank	Category	2025 Market	CAGR	Stage	Core Thesis
1	AI-Native DCIM and Energy Optimization	\$3.6B	18.3% to \$9.9B by 2031	Seed to Series B	Legacy energy management software at the precise moment power is the scarcest resource. No AI-native category leader. Top 5 hold only 26% combined share.

2	Inference Optimization and Multi-Cloud MLOps	\$2.33B	28.9% to \$25.9B by 2034	Series A to B	2025 AI build-out was training-focused. Inference is the next three years. No clear category winner. Hyperscalers only compete within single clouds.
3	Zero Trust and DC Security Automation	\$17.9B	11.9% CAGR	Seed to Series A	AI workloads expanded attack surface. EU AI Act enforcement drives mandatory spends. No incumbent has purpose-built AI compliance modules.

Emerging Whitespaces

Three whitespaces in the software stack deserve priority attention because they are pre-competitive, structurally necessary, and not yet served by any incumbent. Federated multi-site AI-native DCIM does not exist as a commercial product today. Current DCIM tools are single-site and hardware-vendor-specific. Distributed AI infrastructure, combining GPU clusters with edge inference nodes and colocation-hyperscaler hybrid deployments, requires a federated DCIM that provides a unified control plane across physically disparate facilities. No vendor has built this.

Energy-aware workload orchestration, meaning software that integrates GPU cluster scheduling with real-time energy market signals and carbon accounting, is another open category. A training run scheduled around real-time electricity pricing offers significant cost savings at scale. No company has combined compute scheduling, energy market participation, and carbon accounting into a single platform.

AI regulatory compliance automation is the strongest whitespace on risk-reward across the entire software stack. EU AI Act GPAI obligations are already in effect. Full enforcement for high-risk AI begins August 2026. NIST AI RMF is the de facto US procurement requirement. Neither existing GRC platforms nor pure-play tools like Vanta or Drata have purpose-built modules for AI system compliance. This is regulation-driven, non-discretionary, and carries high switching cost once deployed.

Investment Screening Criteria

Criterion	Why It Matters
Solves a problem created or worsened by AI	Ensures the market is growing toward the company, not away from it
Technical moat not replicable in 12 to 18 months	Physical layer integrations, proprietary training data, or network effects create defensibility
Multi-cloud or hardware-agnostic architecture	Single-cloud or single-vendor products face existential hyperscaler substitution risk

Quantifiable ROI measurable within 90 days	Concrete, short-cycle ROI compresses enterprise sales cycles significantly
Net revenue retention above 120 percent	High NRR signals genuine lock-in and validates the LTV/CAC model
Two or more named reference customers at \$100K or more ARR	Validates that a repeatable GTM motion has emerged beyond pilot stage
Regulatory tailwind backing the category	Compliance-driven demand is non-discretionary and carries structurally higher retention
Founder team with infrastructure expertise and enterprise sales experience	Long data center software sales cycles require both dimensions to perform on GTM
Not on hyperscaler roadmap within 24 months	A category AWS, Azure, or GCP is actively building is a value trap regardless of current growth rate

SECTION 6 | CROSS-CUTTING THEMES AND SYNTHESIS

Where the Five Research Streams Converge

Five researchers, five domains, and one consistent finding: the infrastructure constraints binding AI buildout today are not demand problems. They are physics problems, architecture problems, and supply chain problems, each with specific technologies and companies addressing them. The question for ClearVision Ventures is not whether these problems get solved. Market forces, hyperscaler capital expenditure, and regulatory pressure make resolution structurally inevitable. The question is which companies capture the value when solutions reach commercial scale.

Energy as the Central Constraint

Every research stream in this report touches power. GaN eliminates conversion losses inside the data center. HTS eliminates transmission losses outside it. Photonic interconnects reduce the energy cost of moving data by over 90 percent. GPU orchestration software recovers usable capacity from idle hardware without adding a watt of power. AI-native DCIM manages power and cooling as integrated, real-time optimization problems. Waste-heat recovery converts thermal output into a revenue stream rather than an environmental liability.

The IEA projects 945 TWh of data center electricity consumption by 2030. Data center power demand is forecast to grow 165 percent by 2030 versus 2023 levels (Goldman Sachs Research, February 2025). Morgan Stanley projects a 49 GW US power shortfall by 2028 that cannot be built away in time. The companies that relieve power constraints are not addressing a niche. They are addressing the binding constraint on the growth of the entire AI infrastructure buildout.

The Regulatory Tailwind Is Real and Accelerating

All five research streams identified regulatory catalysts that are already active rather than merely anticipated. EU Energy Efficiency Directive obligations entered force in January 2025. Germany's 20 percent waste-heat

reuse requirement has an enforcement date attached to it. EU AI Act GPAI obligations are in effect, with full enforcement for high-risk AI beginning August 2026. NIS2 Directive enforcement is ongoing. US EPA reporting mandates are creating compliance infrastructure demand across the data center sector.

Regulation-driven demand is structurally different from market-driven demand. Operators must respond regardless of budget cycles or procurement timelines. Retention is higher because compliance software is not discretionary. Switching costs are elevated because replacing a deployed compliance system creates regulatory risk during the transition. For ClearVision's portfolio construction, regulatory tailwinds convert what might otherwise be discretionary purchases into non-discretionary ones, which changes the underwriting assumptions in a meaningful way.

The Investment Window Is Closing, Not Opening

Across all five domains, the narrative is the same: commercial validation has happened or is happening now, but early-stage valuations remain accessible for a limited time. Celestial AI's \$3.25 billion exit validates photonic chip design. NVIDIA's \$700 million acquisition of Run:AI validates GPU orchestration software. Baseten's \$75 million Series C and Lambda's \$1.5 billion Series E validate inference infrastructure. VEIR's oversubscribed \$75 million Series B validates HTS power delivery. Ecolab's \$4.75 billion CoolIT acquisition validates liquid cooling hardware.

These are not signals that the window is open. They are signals that it is compressing. By the time Microsoft Azure announces large-scale HTS deployment, VEIR's early-stage valuation access will be gone. By the time photonic interconnects are in 30 percent of tier-1 hyperscaler clusters, the Ayar Labs and POET Technologies rounds at venture valuations will be done. The thesis is not five years out. It is this year and into 2027.

Consolidated Investment Recommendations

The table below brings together the priority investment opportunities identified across all five research domains. It is intended as a single-page reference for ClearVision's investment team when evaluating pipeline against the themes in this report.

Domain	Company or Category	Stage	Conviction	Check Size	Key Catalyst or Deadline
Power: GaN	MTM Critical Metals	Early stage	Very High	\$3M to \$8M	Nov 27, 2026 – China gallium ban expiry
Power: GaN	Amber Semiconductor	Series C	High	\$5M to \$15M	Q3 2026 – PowerTile hyperscaler shipments begin
Power: HTS	MetOx International	Series B Extension	Very High	\$5M to \$15M	Now – pre-revenue, post-DOE validation entry
Power: HTS	VEIR	Series B/C	High	Co-invest	2027 – commercial launch window closing
Power: HTS	SuperNode	Pre-commercial	Medium	\$3M to \$8M	Now – early-relationship before Series B
Cooling	Waste-heat recovery (Karman, Qpinch)	Series A/B	Highest	\$5M to \$15M	Now – EU EED compliance mandatory, Jul 2028
Cooling	GPU orchestration software (Crusoe, Giga ML)	Seed to Series A	Very High	\$3M to \$10M	Now – tokens per watt metric accelerating
Cooling	Two-phase liquid cooling (Accelsius, Submer)	Series B/C	High	\$10M to \$20M	2026 to 2027 – Vera Rubin deployments forcing transition
Cooling	AI-native DCIM platforms	Series A Lead	Good	\$5M to \$12M	Source now – category forming in real time
Silicon	Vendor-neutral AI chip optimization	Seed to Series A	High	\$3M to \$10M	Now – inference cost pressures acute across operators
Photonics	Photonic chip design (Lightmatter, Ayar Labs)	Series A/B	Very High	\$5M to \$15M	2026 to 2027 – UALink ecosystem opens commercially

Photonics	POET Technologies	Growth stage	High	\$3M to \$10M	H2 2026 – 3.2T optical engine sampling begins
Software	AI-native DCIM and energy management	Seed to Series B	Very High	\$3M to \$12M	Now – EU EED and EPA mandates converting demand
Software	Inference optimization and multi-cloud MLOps	Series A to B	Very High	\$5M to \$15M	2026 to 2027 – 12-18 month Series A window open
Software	AI regulatory compliance automation	Seed to Series A	High	\$3M to \$10M	Aug 2026 – EU AI Act full enforcement begins

Conclusion and Strategic Recommendations

Eleven and a half weeks of concurrent research across five domains of AI data center infrastructure leads to a clear directional view, though confidence levels differ between technologies and across the investment profiles identified in each section.

The power infrastructure thesis carries the highest near-term conviction available to ClearVision. GaN is already deployed in production AI servers, has a clear architectural catalyst in the 800V HVDC transition, and has a known supply chain risk date in the November 27, 2026 gallium export ban suspension expiry. HTS is building commercial momentum through Microsoft, VEIR, and MetOx, and the pilot-to-contract conversion window of 2026 to 2027 is the last early-stage entry point before valuations reflect commercial revenue. The pick-and-shovel plays, MetOx for HTS wire and MTM Critical Metals for domestic gallium, offer the best risk-adjusted entries in the power category.

The cooling and thermal management thesis is best deployed across a two-speed strategy. Waste-heat recovery and GPU orchestration software should be funded now: regulatory tailwinds are already in force, economics are improving independently, and the Series A/B entry stage is active. Two-phase liquid cooling and advanced thermal materials are medium-term positions where the proof-of-concept-to-commercial-scale conversion happens between 2027 and 2030, with strategic industrials like Johnson Controls and Legrand providing distribution validation and co-investment anchoring.

The silicon efficiency thesis is cleaner conceptually than it is in execution. The category is real: vendor-neutral AI chip optimization infrastructure that improves inference economics, GPU utilization, and workload orchestration across heterogeneous fleets is a genuine gap in the market. The challenge is sourcing teams with the right combination of systems depth and enterprise sales experience, and underwriting only when technical validation and commercial evidence are both present. The recommended approach is to formalize a silicon efficiency thesis, build a focused pipeline at seed to Series A, and hold to a high evidence bar before deploying capital.

The photonic interconnect thesis has the clearest exit validation of any category in this report through the \$3.25 billion Celestial AI acquisition. Investment should concentrate on the three layers with highest venture fit: photonic chip design, advanced packaging and integration, and laser materials innovation. Cross-ecosystem compatibility with both NVLink and UALink is a minimum requirement for any chip design company. Laser supply security through multi-year indium phosphide agreements is a condition of investment for any photonics portfolio company. The UALink ecosystem opening in late 2026 and 2027 is the seed-stage monitoring priority for this domain.

The software thesis is perhaps the most immediately actionable for a fund at ClearVision's check size and stage. AI-native DCIM and energy optimization sits at Seed to Series B with a massive fragmented incumbent landscape and mandatory regulatory demand. Inference optimization and multi-cloud MLOps sits at Series A to B with 28.9 percent CAGR and no clear category winner. AI compliance automation is pre-competitive, regulation-driven, and has zero incumbent solutions. These three categories represent the sourcing priority order for the data center software mandate.

Taken together, the five research streams point toward a single organizing principle for ClearVision's AI infrastructure mandate: the best returns will come from software and enabling technologies where AI has

made existing tools no longer good enough. The hardware will get built. The compute will get deployed. The money is already committed. What is missing is the layer that makes the hardware work efficiently, safely, compliantly, and sustainably. That is where ClearVision has an edge, and where the window for early-stage entry at venture valuations remains open for months rather than years.

Appendices

Appendix A: Key Market Data Summary

Market Segment	2025 Size	Projected Size	CAGR	Source
Power GaN Devices	\$355M	~\$3B by 2030	42%	Yole Group 2025
DCIM Software	\$3.6B	\$9.9B by 2031	18.3%	Mordor Intelligence 2025
MLOps Platforms	\$2.33B	\$25.9B by 2034	28.9%	Fortune Business Insights 2025
Data Center Security	\$17.9B	Not yet sized	11.9%	Precedence Research 2025
IT Operations Management (ITOM)	—	\$81B by 2028	10.3%	Gartner 2024
Software-Defined Data Center	\$72.3B	Not sized	30.4% to 2033	Grand View Research 2024
Scale-Up Interconnect	—	\$36B+ by 2030	Not disclosed	650 Group Jan 2026
Data Center AI Market	\$242B	\$1.2T by 2030	~5x growth	McKinsey 2025

Appendix B: Dated Catalysts and Investment Timeline

Period	Catalyst	Domain	Investment Implication
Now to Q3 2026	NVIDIA Vera Rubin volume shipments begin. AWS, Google, and Microsoft are first deployers.	GaN	800V HVDC adoption locking in for the next hardware generation.
Q3 2026	Amber Semiconductor PowerTile shipping to hyperscaler customers.	GaN Application	Live entry window with named delivery milestone.
Late 2026	GlobalFoundries and Navitas domestic GaN manufacturing online.	GaN Supply	US supply chain independence from China takes material step forward.
November 27, 2026	China gallium export ban suspension expires.	GaN Supply	MTM Critical Metals structural moat appreciates. Reinstatement risk becomes live.

Late 2026	UALink ecosystem opens commercially.	Photonics	Seed-stage entry window for UALink-compatible photonic engine companies.
2026 to 2028	Microsoft HTS pilot validation data from VEIR deployments.	HTS	Binary validation moment for VEIR and MetOx. Defines Series C entry thesis.
2027	VEIR full commercial launch. NVIDIA Vera Rubin HVDC architecture formalized.	Both	Last early-stage valuation access window before commercial revenue reprices the category.
August 2026	EU AI Act full enforcement for high-risk AI systems begins.	Software	AI compliance automation becomes mandatory for enterprise operators.
2027 to 2030	REBCO tape costs approach \$10 per meter target via fusion capex spillover.	HTS	Economics of data center HTS deployment reach commercial viability threshold.
2029	Vertiv projects 1 MW per rack in leading AI deployments.	Both	Forces next-generation power architecture across the entire industry.

Appendix C: Investment Screening Checklist

The following criteria apply as a unified screening framework across all five domains. Criteria 1 through 4 are minimum thresholds. Criteria 5 through 9 differentiate between candidates that pass the threshold.

No.	Criterion	Minimum Threshold
1	Solves a problem created or acutely worsened by AI	Must articulate how AI workload growth directly expands TAM
2	Technical moat not replicable in 12 to 18 months	Physical layer integrations, proprietary training data, patents, or network effects required
3	Multi-cloud or hardware-agnostic architecture	Single-cloud or single-vendor products face existential hyperscaler substitution risk
4	Quantifiable ROI measurable within 90 days	Customer must be able to measure cost reduction within a fiscal quarter

5	Net revenue retention above 120 percent	Validates genuine lock-in and LTV/CAC model for VC underwriting
6	Two or more named reference customers at \$100K or more ARR	Must have crossed from pilot to production with a repeatable GTM motion emerging
7	Regulatory tailwind backing the category	Compliance-driven demand carries structurally higher retention than discretionary spend
8	Founder team with infrastructure expertise and enterprise sales	Both dimensions required. Teams without enterprise sales consistently underperform on GTM.
9	Not on hyperscaler roadmap within 24 months	A category AWS, Azure, or GCP is actively building is a value trap regardless of current growth rate

Appendix D: Sources and References

Power Infrastructure: GaN

1. IISc Bangalore. AlTiO/p-GaN gate stack. IEEE Transactions on Electron Devices, 2025.
2. Navitas Semiconductor. 10 kW DC-DC Platform Press Release. February 9, 2026.
3. Navitas Semiconductor. 800V-to-6V Power Delivery Board Debut at NVIDIA GTC 2026. March 16, 2026.
4. Texas Instruments. 800V DC Power Architecture Unveiled at GTC 2026. March 16, 2026.
5. Navitas Semiconductor and GlobalFoundries. Strategic Manufacturing Partnership Announcement. 2025; US availability late 2026.
6. Infineon Technologies. 300mm GaN Wafer Production Launch. 2024.
7. Renesas Electronics. Gen IV Plus GaN FETs for AI Data Centers. July 2025.
8. GlobalFoundries and EDN. How HVDC and GaN Are Transforming Hyperscale Data Centers. February 2026.
9. Efficient Power Conversion (EPC). APEC 2026 CEO Interview.
10. Yole Group. Power GaN Market Report 2025.
11. onsemi. Vertical GaN Semiconductors Announcement. March 2026.
12. Intel Foundry Services. World's Thinnest GaN Chiplet Announcement. April 9, 2026.
13. Atomera and Synopsys. GaN Device Development Collaboration Expansion. April 23, 2026.
14. Innoscience. Critical Design-Ins for Google AI Hardware Platforms. 2025.
15. Navitas Semiconductor. ISO 14040/14044 Lifecycle Assessment. Certified.

Power Infrastructure: HTS

16. University of Houston and PNAS. 151K ambient-pressure superconductivity record. March 9, 2026.
17. Chalmers University of Technology. Boosting superconductivity in ultrathin YBa₂Cu₃O₇ films via nanofaceted substrates. Nature Communications, March 2026.

-
18. VEIR and Microsoft. 3MW HTS Cable Demonstration. OCP 2025 Summit, November 2025.
 19. Microsoft Azure Engineering Blog. HTS Datacenter Power Overview. February 10, 2026.
 20. IEEE Spectrum. AI Data Centers Study High-Temperature Superconductors. February 26, 2026.
 21. MetOx International. Beyond Voltage: Rethinking Data Center Density with HTS. Data Center Dynamics, September 24, 2025.
 22. 2025 Superconductivity for Data Centers Workshop. Industry participation summary.
 23. KIT AmpaCity, Essen. IEEE Documentation of 1km HTS Cable in operation since 2014.
 24. Wood Mackenzie. Copper Deficit Report. 2025.
 25. Morgan Stanley. US Power Shortfall Projection to 2028.
 26. Gartner. AI Data Center Power Constraint Estimate. November 2024.

Cooling and Thermal Management

27. International Energy Agency. World Energy Outlook 2025.
28. Lawrence Berkeley National Laboratory. Data Center Energy Report 2025.
29. S&P Global and 451 Research. Data Center Market Analysis. November 2025.
30. Goldman Sachs Research. Data Center Power and Energy Software. February 2025.
31. ABI Research. AI Infrastructure and Cooling Markets. January 2026.
32. Uptime Institute. Annual Global Data Center Survey 2024 and 2025.
33. Synergy Research Group. Data Center Market Share Analysis. 2025.
34. Vertiv Holdings. 8-K Filing. February 2026.
35. Ecolab. 8-K Filing, CoolIT Acquisition Announcement. March 2026.
36. Luccioni, Strubell, and Crawford. Power Hungry Processing. ACM FAccT 2025.
37. arXiv preprint 2602.05116. AI Energy and Compute Scaling. February 2026.
38. ARPA-E COOLERCHIPS Programme. Technical documentation and programme goals.

Silicon Efficiency

39. International Energy Agency. Energy and AI Programme Report. 2025.
40. International Energy Agency. Electricity Mid-Year Update 2025.
41. Microsoft. Maia 200 AI Accelerator Release Documentation. 2025.
42. NVIDIA. Spectrum-X Photonics Announcement. 2025.
43. Google Cloud. TPU v7 Ironwood Documentation. 2025.
44. NVIDIA. Run:AI Acquisition Disclosures. 2023.
45. Fortune Business Insights. MLOps Market Report. 2025.
46. Grand View Research. Container Orchestration Market Analysis. 2024.

Photonic Interconnects

47. International Energy Agency. Data Center and AI Energy Outlook. April 2025.

-
48. McKinsey Global Institute. AI Infrastructure and Data Center Investment. April and July 2025.
 49. Goldman Sachs Research. Data Center Networking and Photonics. November 2025.
 50. Yole Group. Photonic Interconnects Market Report. 2025.
 51. IDTechEx. Photonics and Data Center Infrastructure. 2026.
 52. 650 Group. Scale-Up Interconnect Market Forecast. January 2026.
 53. TrendForce. Co-Packaged Optics Market Analysis. December 2025.
 54. Semiconductor Engineering. Optical Fiber Communications Conference Report. April 7, 2026.
 55. Broadcom. Co-Packaged Optics Platform Launch. October 2025.
 56. Marvell Technology. Celestial AI Acquisition Announcement. February 2026.
 57. UALink Consortium. Version 2.0 Interconnect Standard Release. April 7, 2026.
 58. HPCwire. Co-Packaged Optics and AI Cluster Design. December 2025.

Data Center Software

59. Dell'Oro Group. Data Center Capex and Software Pull-Through. March 2026.
60. Grand View Research. Software-Defined Data Center Market Analysis. 2024.
61. Gartner. IT Operations Management Spending Projection. 2024.
62. Global Market Insights. DCIM Market Share and Competitive Landscape. 2024.
63. Mordor Intelligence. DCIM Market Size and Forecast. 2025.
64. Fortune Business Insights. MLOps Platform Market. 2025.
65. Precedence Research. Data Center Security Market. 2025.
66. Datadog. Annual Report and 10-K Filing. FY2025.
67. Cisco. Splunk Acquisition Press Release. March 2024.
68. BusinessWire. Baseten Series C Funding Announcement. February 2025.
69. Lambda. Series E Funding Announcement. November 2025.
70. European Union. EU AI Act Official Timeline and GPAI Obligations. 2024.
71. Energize Capital. Data Center Efficiency Savings Analysis. 2025.

This report synthesizes secondary sources published between 2024 and April 2026. No primary interviews, original surveys, or empirical data collection were conducted.

Confidential | For Internal Use Only | ClearVision Ventures | April 2026

